

A Haptic Navigation Aid for Individuals with Visual Impairments: Indoor and Outdoor Feasibility Evaluations of the LaserNavigator

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Navigation is largely mediated by vision. Thus, visual impairments (i.e., blindness or low vision) limit this ability (Brouwer, Sadlo, Winding, & Hanneman, 2008), which can lead to a decreased quality of life (Nyman, Dibb, Bridget, & Gosney, 2012). The white cane is an excellent solution for detection of obstacles that are near the ground and in close proximity to the traveler, but there is a lack of feasible options for larger ranges. The few commercial products that provide solutions for larger ranges have not had a wide impact as they are not frequently utilized by people with visual impairments (Pey, Nzegwu, & Dooley, 2007), thus making further innovation and research important.

Electronic mobility aids for people with visual impairments are commonly divided into electronic orientation aids (EOAs) or electronic travel aids (ETAs). EOAs include GPS-based technology, while ETAs include a measurement system (e.g., ultrasound or laser), to identify objects and obstacles. In a previous study, Innala Ahlmark, Prellwitz, Röding, Nyberg, and Hyyppä (2015) presented a laser-based technology combined with haptic feedback. The device was

attached to a table on wheels due to its size and weight. However, a handheld solution is necessary for applicability in daily activities.

This research report presents two separate evaluations, one indoor and one outdoor experiment, of a novel handheld mobility aid (the LaserNavigator) that uses a laser to measure the distance to objects. Since it is an optical system, rather than an ultrasonic system as is often utilized, it can measure large distances with high accuracy and a small beam spread, which is of importance in this case since one intended application is to determine direction and distance of specific landmarks such as door openings, buildings,

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and other objects. The aim of the research presented here was to commence an initial evaluation of the usability of the LaserNavigator, to identify movement patterns and strategies when using the device, and to understand users' perceptions of usability of the device.

Method

A mixed-methods approach was used in this two-phase study. In Phase 1 (indoor evaluation), objective assessment with a motion-capture system was combined with visual observation of movement behavior via video recordings, followed by a systematic interview. In Phase 2 (outdoor evaluation), which occurred 4 months after Phase 1, visual observation via video recording was combined with a systematic interview. Ethical approval was given by the regional ethical review board, Umeå, Sweden (reference 2010-139-31) and informed consent was obtained from all participants.

PARTICIPANTS

The participants were recruited through the local district of the Swedish National Association of the Visually Impaired. Inclusion criteria were age 18 years or older, blind, and able to walk and hold the LaserNavigator. Three participants Anna, Birgitta, and Carl (pseudonyms); two females and one male; aged 60, 72, and 78 years, respectively, were included in both Phase 1 and 2 evaluations of the LaserNavigator. Anna and Birgitta were independent travelers, while Carl did not travel outdoors without a guide. All subjects were diagnosed as blind according to the *International Statistical Classification of Diseases and Related Health Problems*, 10th revision (ICD-10; World Health Organization, 2016). Anna was in Category 4 (light perception) and Birgitta and Carl were in Category 5 (no light perception). Anna



Figure 1. A photo of the LaserNavigator, showing the laser rangefinder (1), ultrasound sensor (2), and the loudspeaker (3).

was congenitally blind, while Birgitta and Carl were adventitiously blind, and they had been blind for 2 and 7 years, respectively. All three used white canes, but Carl used it only as a walking cane. Additionally, Birgitta had a dog guide and used a GPS device daily.

EQUIPMENT

The LaserNavigator (Figure 1) used a laser range finder to measure the distance. The laser measures distance up to 50 m with high accuracy (error less than 10 cm + 1% of range) and with a beam spread of 0.2°. Haptic feedback was supplied through the index finger placed on top of a small vibrating loudspeaker. When the object was at or closer than the selected distance (virtual “cane length”), the speaker membrane emitted short vibration pulses. To vary the “cane length,” the user moved the LaserNavigator closer to or further away from their body. The device possessed an ultrasound sensor determining the body–device distance. This value is then multiplied by a constant factor to set the “cane length.” The system functions as a virtual telescopic white cane that automatically expands or contracts depending on how far away from the body

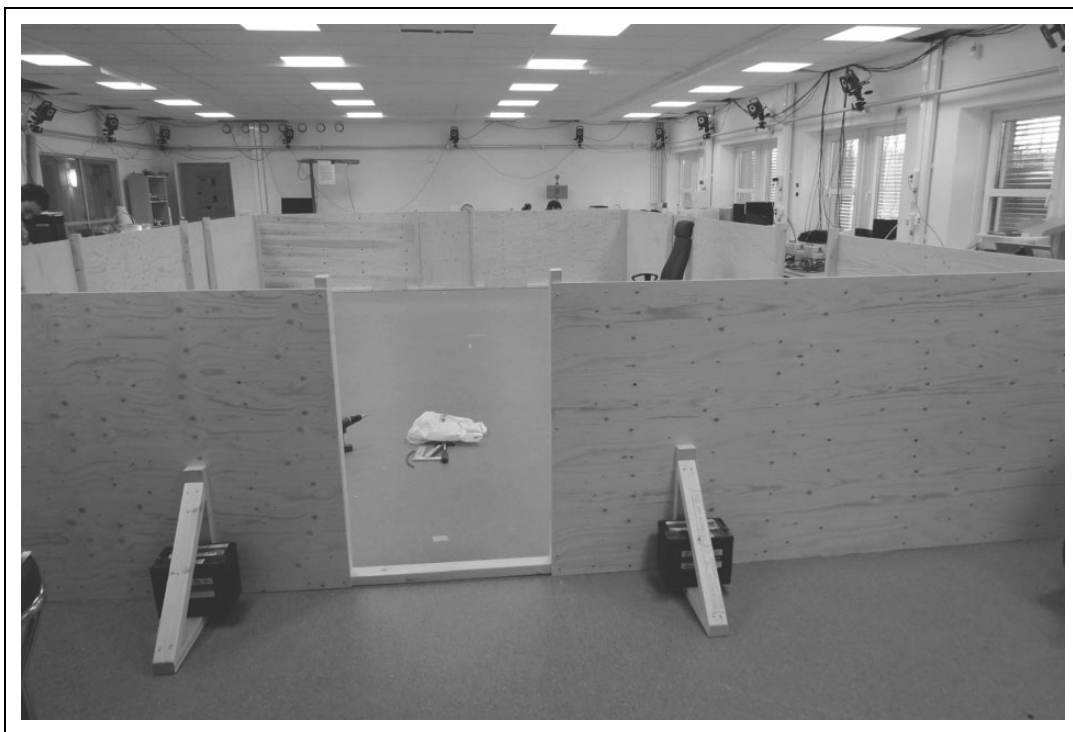


Figure 2. Picture of the makeshift room used for Phase 1, as viewed from outside the entrance door.

the user holds it. In the indoor trials, the scale factor was set to 10, so that a body–device distance of 50 cm would equate to having a 5-m-long “virtual cane,” while in the outdoor trials, the scale factor was 50. The weight of the LaserNavigator was 300 g.

TEST ENVIRONMENTS

In Phase 1, the tests were performed in a motion laboratory with 20 motion-capture cameras. A makeshift room of the size 5.6 m by 8.8 m (Figure 2) was constructed of plywood with a wall height of 1.2 m. The shorter sides of the room had one door while the longer had two. The low walls were detectable with the LaserNavigator, while still allowing the motion-capture cameras to track markers moving about inside the room. Reflector markers were placed on the participant and the LaserNavigator. In Phase 2,

the tests were performed outdoors in an area adjunct to the university.

TASKS

The participants had 30–60 minutes to familiarize themselves and train with the LaserNavigator before each test phase

Phase 1 included the following steps:

- (1) The participants were positioned at a predetermined starting position outside the makeshift room approximately 4 m from the entrance door.
- (2) They then located the entrance door (same door each time) and entered the room.
- (3) Next, they needed to locate another randomly opened door and move to that opening.
- (4) Finally, they needed to find their way back to the entrance door.

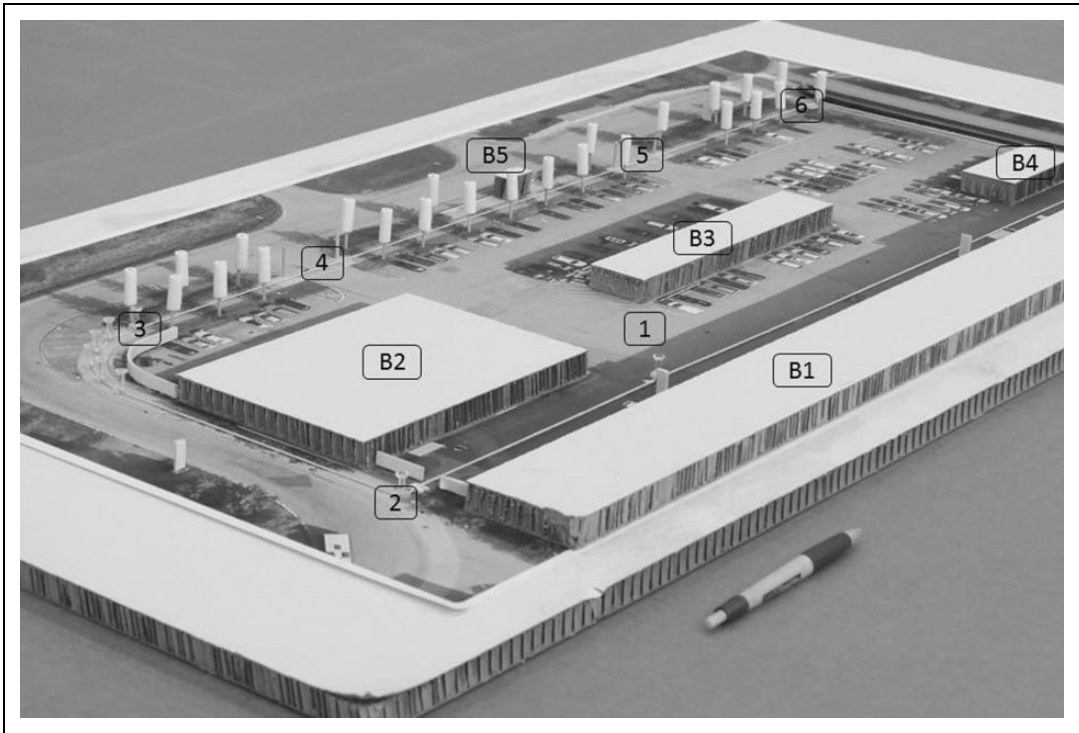


Figure 3. The tactile model to familiarize with the route in Phase 2 of the trial. The route starts at (1) and is represented by a thread. Using the walls of buildings (B1) and (B2) as references, the participants walked toward stairs (2) lined by a fence. Turning 90° to the right and continuing, following the wall of building (B2), the next point of interest was at (3). Here, another fence on the right side could be used as a reference when taking the soft 90° turn. The path from (3) to (6) is through an alley lined with trees. Along this path, the participants encountered crossings (4) and (5), in addition to the bus stop (B5). At (6), a snowdrift guided the participants into the next 90° turn. Building (B4) was the cue to perform another turn and then walk straight back to the starting point (1), located just past the end of (B3).

These steps were repeated 3 times. This design allowed for both predictable and unpredictable elements within the task. Finding the entrance door from the starting position can be considered relatively predictable after a few practice trials. The next part, finding a randomly opened door, can be considered more unpredictable. The last part, finding the way back to the entrance door, can be considered either predictable or unpredictable depending on one's sense of location inside the room.

In Phase 2, the task was to walk a 385-m path among a cluster of buildings, repeated

twice with a 15-minute break in between. The route contained different landmarks (e.g., buildings and trees). Prior to the tests, the participants explored a haptic model of the outdoor environment (Figure 3) to familiarize with the environment. One researcher guided and instructed the participants on how to use and point the LaserNavigator during the trials.

OBSERVATIONS

In Phase 1, all trials were recorded by the motion-capture cameras at 100 frames per

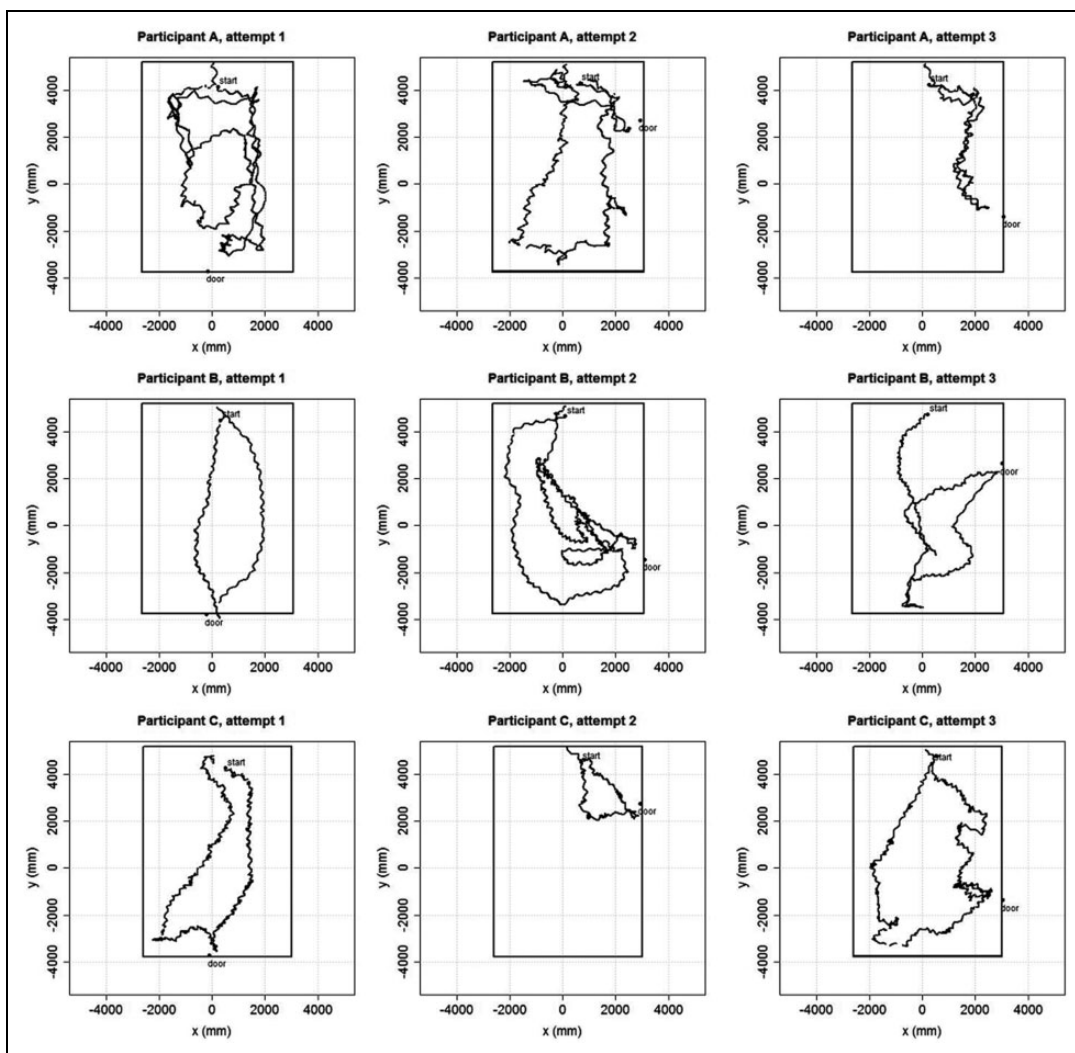


Figure 4. Movement traces recorded by the motion-capture system in Phase I for each participant and their three trials.

second. The recordings were plotted for visual analysis of movement behavior and strategies including the walking trace (Figure 4) and hand movement of the LaserNavigator (Figure 5).

In Phases 1 and 2, all trials were video filmed. The video recordings were visually analyzed independently by three of the researchers, and key points regarding movement patterns and strategies were

summarized. Discrepancies were discussed among the researchers.

INTERVIEWS

After each experiment, the participants were interviewed based on a semistructured 10-question interview guide. The interview guide included questions about the experience of using the LaserNavigator; asking

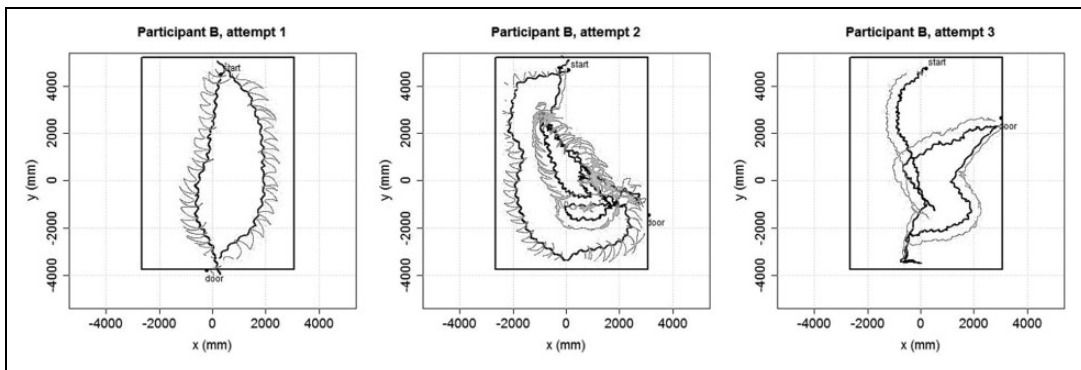


Figure 5. The figure shows the three attempts of Birgitta, with the additional gray line indicating the position of the LaserNavigator. A reduced movement amplitude of the hand and arm movement is shown for attempt three, probably due to fatigue from holding and moving the LaserNavigator.

participants about its usefulness, feasibility, contribution to their perception of the environment, specific occasions when it could be useful, improvements, and their current strategies when losing orientation. The interviews were transcribed verbatim and analyzed based on content analysis (Graneheim & Lundman, 2004). To enhance trustworthiness, the interviews were read and analyzed independently by two of the authors and then discussed to finalize the categories.

Results

MOVEMENT BEHAVIOR

In Phase 1, Anna and Birgitta brought their white canes during the trials. Carl did not bring a cane and was instead guided by one of the researchers. However, all participants used only the LaserNavigator to find the door openings. Combining the back-and-forth movements with the sweeping motion to “touch the wall” in the search for a door opening was generally difficult. All participants, however, learned relatively quickly to find the entrance door from the starting point. The movement traces inside the makeshift room for each participant and each trial are depicted in Figure 4.

In Phase 2, all participants brought their white canes but did not use them much, instead they mainly used the LaserNavigator. They used the device to find objects, typically a “corridor,” that is, an open path straight ahead with “walls,” for example, buildings, on both sides. When they found this corridor, they seemed confident in walking through it by following one or both of the “walls.”

All participants had difficulties holding the LaserNavigator horizontally, and they often needed instructions to angle the device up or down in both the indoor and outdoor tests.

INTERVIEWS

The analysis of the interviews resulted in three categories: benefits, challenges, and future possibilities. The generalizations below are illustrated with quotations from the participants.

Benefits

In general, the LaserNavigator was considered more beneficial in the outdoor environment compared to indoor use. The participants reported that it seemed easier to walk in a

straight line and to walk more freely without having to touch something with the white cane in the outdoor environment. One participant described it this way: "I always have to have something to follow; now I could feel both sides and walk in the middle." Another positive experience was the property of being able to greatly vary the "cane length."

Challenges

The general opinion was that the tasks were challenging and it was difficult to hold the device horizontally. In particular, Birgitta felt using the device was tiring due to its weight and saw difficulties combining it with the white cane, a GPS, and a dog guide. Even though the device was considered more beneficial for the outdoor environment, this setting was also seen as more challenging. Unfamiliar environments and tasks were more difficult compared to those that were familiar. The haptic model of the outdoor environment was useful. This can be exemplified by a quote from one of the participants: "I memorized it and had it in mind while I walked."

Future possibilities

All participants had suggestions for improvement and described situations where they thought the device would be useful, such as going out of the house to the garden. Going out was easy, but finding the way back was difficult. Similar scenarios included finding the door to a shopping mall or finding a landmark such as a building or a lamppost. In particular, one participant wished for the option to filter objects so only the things of interest would be detectable: "There are so many details outside: bikes, waste bins, trees, other people. . . . [Y]ou might want to find that particular lamppost in all that mess." Carl had several suggestions including mounting the device on a walking cane,

auditory feedback through a headset, a push button making the LaserNavigator vocally announce the exact distance and compass direction, and feedback to help with holding the device horizontally.

Discussion

These first evaluations of the LaserNavigator have contributed important initial results. They have shown that the device can contribute information about the surroundings (e.g., finding doorways in an indoor environment and surrounding objects when walking a specified outdoor route). It was easier to use in a more predictable and familiar situation, which is exemplified by the fact that all three participants quickly learned to find and navigate through the entrance door from the standardized starting point and to identify open space between known buildings.

Several challenges, but also possibilities for improvements, were revealed. Two major difficulties were holding the LaserNavigator horizontally and identifying the distance to objects. Difficulties with range perception have been reported in a previous study with a similar technology (Innala Ahlmark et al., 2015). These difficulties may improve with more practice, while a technical solution could be to include specific audible or tactile feedback. Experience of fatigue during sustained use can be related to the weight and design of the device. Future development should therefore also include weight reduction and design to reduce physical strain and effort.

Limitations of our study include the small number of participants and only including senior citizens. Moreover, the training periods were short, and tests were performed only in unfamiliar environments. The strengths of the study include having only participants who were blind and therefore potential users of the device, and that evaluation was performed in both indoor and

outdoor environments by the same participants. The practical implication is that the LaserNavigator, with some modifications and a prolonged training period, may be a useful secondary mobility device in conjunction with the white cane and is probably more useful in an outdoor environment compared to indoor. Optimal environment (outdoor versus indoor), however, needs to be evaluated in future, larger studies.

Declaration of conflicting interests

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